Proceedings of 2025 IEEE
International Conference on Robotics and Biomimetics
December 3 - 7, 2025, Chengdu, China

2025 IEEE International Conference on Robotics and Biomimetics (ROBIO) | 979-8-3315-5747-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ROBIO66223.2025.11377340

HydroGrasp: A Hydraulic Multi-Fingered Gripper

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***Abstract*— Robotic manipulators need to balance structural lightness with high payload capacity. Conventional motor-in-place designs increase distal inertia and are constrained in power output due to limited space, while cable-driven systems reduce distal mass but suffer from mechanical complexity and poor durability. To address these limitations, we present HydroGrasp, a novel gripper that integrates advantages of both approaches through hybrid electrical-hydraulic actuation. By relocating the motors to the hand's base and transmitting force hydraulically, HydroGrasp achieves the low distal inertia of cable-driven systems while maintaining the robustness and controllability of motor-driven designs. Its reconfigurable architecture enables dynamic switching between position, force, and enhanced force-control modes to accommodate diverse task requirements. The hydraulic transmission also provides compliance that improves impact robustness. In addition, we develop a kinematic model to facilitate effective grasp control. Experiments show that HydroGrasp delivers a peak actuation force of 3.38 N and can reliably grasp objects weighing up to 0.675 kg, handling items of varying shapes and materials. The system is low-cost, with an assembly cost of about $300.**

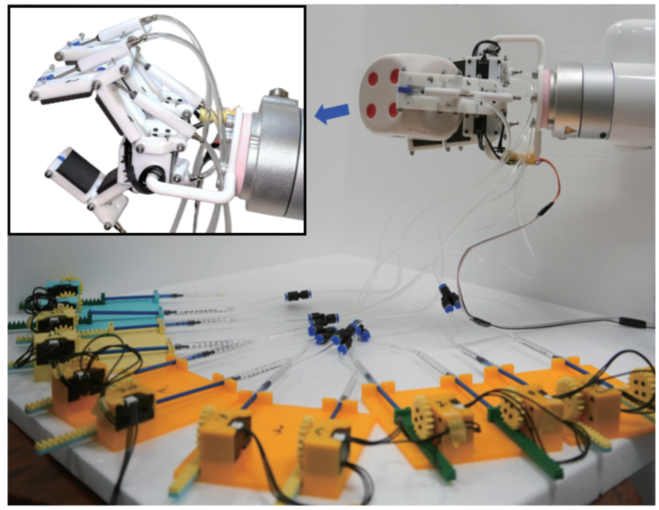

Fig. 1: The HydroGrasp manipulator is mounted on an X-Arm 6 robot and driven by a hydraulic actuator array.

I. Introduction

The human hand is a marvel of biological evolution, offering both exceptional dexterity and payload capacity that surpass most artificially engineered manipulators. It is capable of exerting an average grip force of approximately 450 N, while weighing only around 400 grams [1]. Moreover, it provides over 20 degrees of freedom (DoFs), enabling a wide range of manipulation tasks such as grasping, object reorientation, tool use, and in-hand manipulation [2]. In contrast, many contemporary robotic hands are significantly heavier and deliver only a fraction of the grip force. For instance, the state-of-the-art Shadow Dexterous Hand, despite its high dexterity, weighs over 4 kg and achieves a grip force of only 30–50 N [3]. This large inertia limits their suitability for fine-grained manipulation, particularly in dynamic or delicate tasks. These disparities highlight the need for new actuation strategies that can achieve both low inertia and high output force.

To enhance manipulator's performance, the mechanical design continuously aims to optimize two competing objectives: lightweight design and high payload capacity. A high payload-to-weight ratio enhances a manipulator's dexterity and utility, while a lightweight structure enables faster movement and reduces the mechanical burden on the supporting robotic arm [4]–[6]. To achieve this balance, modern designs have explored strategies such as underactuation and tendon-driven systems. Underactuation simplifies the design by reducing the number of motors but sacrifices independent control over each joint [7]. Tendon-driven systems relocate heavy motors away from the end-effector, lowering distal inertia. However, they introduce mechanical complexity in tendon routing, and the tendons themselves are prone to stretching, friction, and wear, which are all factors that limit maximum transmissible force and reduce long-term reliability [8], [9].

To address these challenges, we explore hydraulic actuation as a viable alternative. Like tendon-driven systems, hydraulic actuators can be remotely located, transmitting power through fluid-filled tubes. A key advantage of hydraulics is their ability to transmit significantly larger forces with minimal loss, without the wear and stretching issues associated with cables. Although traditional hydraulic systems are typically confined to high-power industrial machinery and viewed as complex and leak-prone, its low-power and small-scale versions offer simplicity, robustness, and ease of integration, making them ideal for gripper applications. Previous research has investigated hydraulic solutions for robotic hands [10]–[13], but many are with limitations such as low DoFs, low payload capacity, or fabrication complexity that hinders accessibility for the broader robotics and embod-

This work was supported by Shanghai Frontiers Science Center of Human-centered Artificial Intelligence (ShangHAI), MoE Key Laboratory of Intelligent Perception and Human-Machine Collaboration (KLIP-HuMaCo).

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Project page: https://github.com/m31-1612/HydroGrasp

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ied AI community.

In this paper, we propose HydroGrasp, a 3-fingered robotic gripper as shown in Fig. 1. The HydroGrasp manipulator is designed to be powerful, lightweight, and accessible to the research community. With a self-weight of only 0.36 kg, it achieves a payload capacity of up to 0.675kg through a novel hybrid electrical-hydraulic actuation system. To ensure low cost and ease of fabrication, the hydraulic system is constructed using commercially available components, along with 3D-printed structural parts. We validate HydroGrasp's capabilities through extensive experiments, including force characterization and grasping demonstrations with objects of varying shapes, sizes, and weights, highlighting the gripper's versatility and robustness.

Overall, our contributions are:

1) The design and implementation of HydroGrasp, a 3-fingered, 9-DoF gripper.
2) A low-cost hybrid electrical-hydraulic actuation system for robotic grippers that combines the benefits of remote actuation with high force transmission.
3) Comprehensive evaluations on HydroGrasp's grasping performance.

II. Related Works

A. Robot Manipulator Actuation

The design of robotic manipulators has leveraged various actuation strategies offering distinct trade-offs in force capacity, responsiveness, mechanical complexity, and application suitability. Conventionally, motor-on-joint configurations are among the most prevalent. In these systems, actuators are placed directly at each joint, enabling high-fidelity motion control and feedback that are beneficial for tasks requiring fine dexterity [14], [15]. Nevertheless, this architecture increases the mass and inertia of distal links, reducing energy efficiency and degrading dynamic performance. Besides, actuation power is often constrained by the tight spatial limitations of joint housings, especially in compact or anthropomorphic robots [16]. Examples include the Leap Hand [17] and the DLR Hand System [18], which prioritize precision but impacted by high weight and size penalties at the fingertips.

Tendon-driven architectures offer an alternative approach by transmitting power via flexible tendons routed from motors placed at more proximal locations, such as the forearm or base. This significantly reduces the mass and inertia at the end-effector, enabling dense, multi-DoF designs within compact form factors. Notable implementations include the Shadow Dexterous Hand [3] and the da Vinci Surgical System [19], both of which achieve high dexterity in constrained spaces. However, tendon-based systems present mechanical challenges: complex routing, tension maintenance, and issues such as friction, hysteresis, and backlash can degrade control accuracy and transmission efficiency [16], [20]. Under high loads, tendons may also stretch or wear out, reducing long-term reliability [8], [9], [16].

Fluidic actuation, which includes both pneumatic and hydraulic methods, transmit power through pressurized gas or fluid. Such design offers lightweight, compliant actuation, and compliance well-suited for soft robotics and safe human–robot interaction. Typical design includes the hydraulic soft gripper [21] and Dexco manipulator [4], which offer a high force-to-weight ratio and low end-effector weights. Their scalability and power density make them a compelling option for applications requiring large output force with minimal onboard mass.

B. Hydraulic Actuation in Robotic Hands

Hydraulic actuation is well known for its high power-to-weight ratio, making it suitable for tasks that demand both strength and responsiveness. Traditionally, hydraulic systems have been deployed in large-scale robots such as Boston Dynamics' Atlas, which showcases both power and agility through maneuvers like backflips [22]. However, these systems often rely on custom-fabricated components that are costly, difficult to produce, and lack commercially available off-the-shelf parts. In contrast, small-scale hydraulic actuation is far more accessible and has seen growing adoption, particularly within the soft robotics community. Systems such as [4], [21] demonstrate the viability of miniature hydraulics using simple, low-cost components like syringes and low-pressure pumps, while still achieving impressive force outputs relative to their size.

Despite this progress, the use of hydraulic actuation in dexterous, multi-fingered robotic hands remains relatively underexplored compared to electric or tendon-driven approaches. Only a few previous efforts, such as the hydraulic anthropomorphic hand [13], have explored this technical path. Our work, HydroGrasp, bridges this gap by introducing a hybrid hydraulic–electric actuation system that combines the strength of hydraulics with the control flexibility of electric drives. Built entirely from commercially available or 3D printed components, HydroGrasp demonstrates that hydraulic actuation can serve not only as a high-power actuation solution but also as a scalable and replicable platform for dexterous robotic manipulation, with the potential to support both research and practical applications.

III. Methodology

The HydroGrasp system comprises two subsystems: the 3-fingered end-effector, and a remote actuation base with electro-hydraulic actuators. This section details the design of the HydroGrasp's mechanical structure, its kinematic model, and the hydraulic actuation mechanism that enables its motion control capabilities.

A. Manipulator Mechanical Design and Kinematics

The HydroGrasp end-effector is shown in Fig. 2. Overall, the end-effector consists of three identical fingers mounted symmetrically on a central palm. Each finger possesses three DoFs: a rotational base joint (roll) and two subsequent flexion/extension joints.

The roll motion of each finger is directly driven by a Dynamixel XL330-M288-T servo motor, which offers a stall

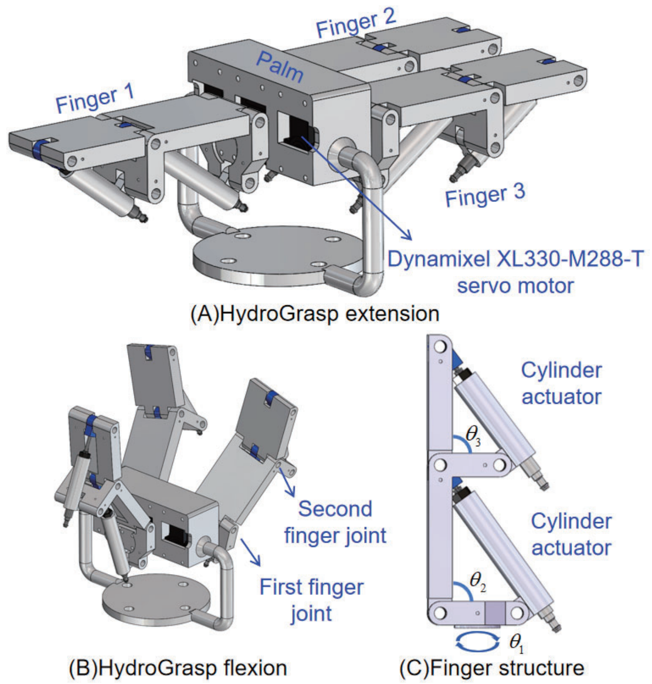

Fig. 2: The structure of the HydroGrasp end-effector. HydroGrasp has three fingers, each with two hydraulically actuated flexion joints and one electrically actuated roll joint at the base.

torque of 0.52 Nm. The roll motor is used for orientation control of the finger's grasping plane. The two flexion/extension joints are each powered by a lightweight, single-acting hydraulic cylinder integrated into the finger's linkage. The linear push motion of these cylinders is converted into angular joint rotation through a triangular linkage mechanism (illustrated in Fig. 2). This design allows the heavy actuators to be located remotely while transmitting high forces to the joints. The current structure achieves a bending range of -0.7° to 75.14° for the first finger joint and -1.57° to 63.19° for the second finger joint.

To model the finger's motion, we define its kinematic chain using the Denavit-Hartenberg (D-H) convention, as detailed in Table I. Here, θ_1 corresponds to the base roll angle, while θ_2 and θ_3 represent the angles of the two flexion joints, which are functions of the hydraulic cylinder displacements.

Based on these parameters, the forward kinematic transformation from the finger base frame to the fingertip frame is given by the matrix $T = T_1^0 \cdot T_2^1 \cdot T_3^2$. The full expression for this transformation is provided in the Appendix.

TABLE I: Denavit-Hartenberg (D-H) parameters for a finger mechanism.

Link	a (mm)	α (rad)	d (mm)	θ(rad)
1	$a_1 = 5$	$-\pi/2$	$d_1 = 12.5$	θ_1
2	$a_2 = 50$	0	0	θ_2
3	$a_3 = 40$	0	0	θ_3

B. Hybrid Electrical-Hydraulic Actuation System

The actuation system for HydroGrasp is shown in Fig. 3. It is housed in a remote base unit and connected to the end-effector via tubing. Each of the two hydraulic DoFs per finger is driven by a syringe linear actuator. The syringe is actuated by a servo motor through a rack-and-pinion mechanism that converts the motor's rotational motion into the linear displacement of the rack. The rack drives the syringe plunger, which transmits pressure through the hydraulic tubing to the cylinders in the finger joints, altering the length of the long side of the triangular structure (Fig. 3 (C)) and enabling flexion and extension of the finger joints.

Our system relies on Pascal's principle for force amplification, using water as the transmission medium. In the transmission of liquid pressure, force transmission ratio is determined by the cross-sectional areas of the syringe (A_s) and cylinder (A_c). The output force at the cylinder, F_c, is given by:

$$F_c = F_s \left(\frac{A_c}{A_s} \right) \tag{1}$$

where F_s is the input force applied by the motor to the syringe. This relationship allows us to generate large output forces at the fingertip using relatively small, low-power motors at the base.

HydroGrasp supports multiple operating modes, enabling it to accommodate a wide range of manipulation tasks. This versatility arises from both the control capabilities of the Dynamixel motors, which support positional and force control, and our proposed dual-drive mechanism that provides an extended range of output force. The dual-drive mechanism is illustrated in Fig. 3 (B). A hydraulic channel employs two servo-driven syringes connected via a Y-connector to a terminal cylinder, generating pressure to control finger motion. This configuration enables the following operating modes:

1) **Position Control Mode:** One servo motor operates in position control mode, while the other syringe is fixed

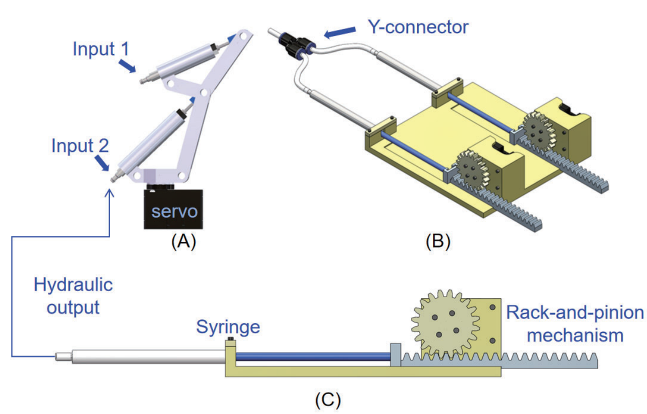

Fig. 3: Hybrid electrical-hydraulic actuation system.
(A) Actuation system of finger. (B-C) Syringe actuator based on rack-and-pinion mechanism.

at its rear end. This mode is used for precise finger positioning to achieve specific poses.

2) **Force Control Mode:** One servo motor operates in current (torque) control mode, while the other is fixed either at the rear end or held in position control mode. This allows HydroGrasp to regulate joint torque and maintain compliance during physical interaction.
3) **Enhanced Force Mode:** Both servo motors drive the syringes synchronously to actuate a single joint, thereby increasing the available output force. The output force can be estimated using the relation:

$$F_{in1}v_{in1} + F_{in2}v_{in2} - P_{friction} = F_{out}v_{out} \quad (2)$$

where $F_{in1} \approx F_{in2}$ are the input forces generated by the motors, $v_{in1} \approx v_{in2}$ are their synchronized velocities, $P_{friction}$ is the power dissipated due to friction, and F_{out} and v_{out} are the net output force and velocity at the terminal output cylinder.

IV. Experiments

To validate the performance of the HydroGrasp manipulator, we conducted a series of quantitative and qualitative experiments. The evaluation was designed to characterize three key aspects of the system: (1) its maximum fingertip force output and payload capacity, (2) the feasibility of its three operating modes, and (3) its versatility in grasping a diverse range of everyday objects.

A. Force and Payload Characterization

First, experiments were conducted to quantify the gripper's maximum fingertip force and payload capacity. To evaluate the maximum fingertip force, the entire three-finger gripper was vertically mounted on the X-Arm 6 robot. Then a single finger was placed to press against a calibrated load cell (Fig. 4 (A)) to measure the output force of the fingertip [23]. For the two joints of the finger, the actuation system for each joint runs in force control mode, with only one servo motor operates in current control mode, while the other locked at the rear end. The driving servo motor was controlled to apply its maximum torque and our HydroGrasp achieved a peak fingertip force of 3.38 N.

To evaluate the payload capacity, a container with screws in it was grasped, as shown in Fig. 4 (B). All servo motors of actuation system operated in enhanced force mode and achieved the highest force output. Screws were incrementally added to the container until the grasp failed (i.e., the object slipped). The mass of the container and its contents just before failure was recorded as the maximum payload. The system demonstrated the ability to safely lift and hold a total payload of 0.675 kg (6.62 N). Compared to the self-weight of 0.36 kg, our HydroGrasp manipulator achieved a force-to-weight ratio of 1.875. Compared with some other robotic hands such as the Shadow Dexterous Hand [3] with a force-to-weight ratio of 1.16, the HydroGrasp manipulator demonstrates remarkable performance among comparable products. Throughout these tests, no leakage was observed in the hydraulic system, which showcased the robustness of our design.

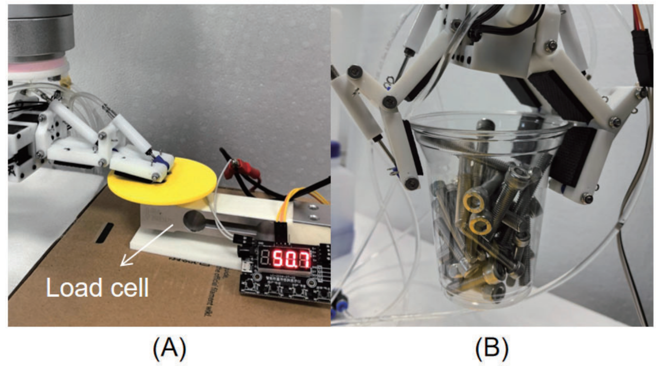

Fig. 4: Experiment setup for characterizing (A) fingertip output force, and (B) manipulator's payload capability.

B. Validation of Three Operating Modes

Next, experiments were conducted to validate the feasibility of the manipulator's three operating modes: position control mode, force control mode, and enhanced force mode. First, the effectiveness of position control was evaluated by analyzing the relationship between servo position commands and the resulting joint angles of the robotic finger. The servo motor was driven with progressively increasing position commands, while the corresponding angular displacements of the two finger joints were recorded simultaneously. Linear regression was then applied to quantify the relationship between input commands and joint responses. The results are shown in Fig. 5. It indicates an approximate linear relationship between the commanded positions and the fingertip motion, showcasing that the finger can be effectively controlled in position mode.

Force control was validated by analyzing the relationship between input current values and the resulting fingertip output force. A single servo motor per joint was driven in current control mode, with the current progressively increased. The output force was measured in real time using a load cell

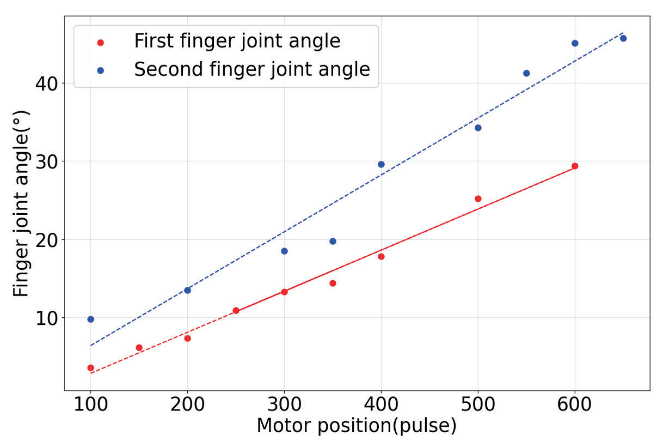

Fig. 5: Relationship between motor encoder position and joint positions in position control mode. (4096 pulse per rotation)

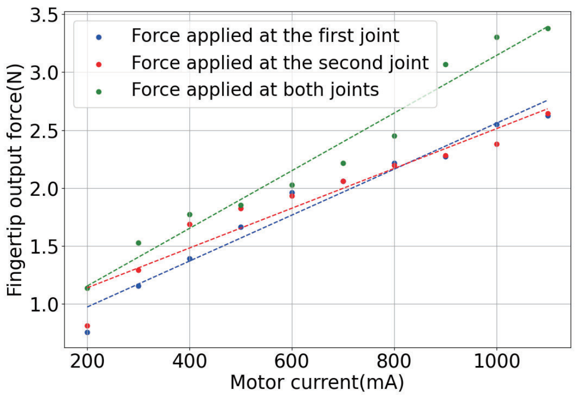

Fig. 6: Relationship between input current value and fingertip output force in force control mode.

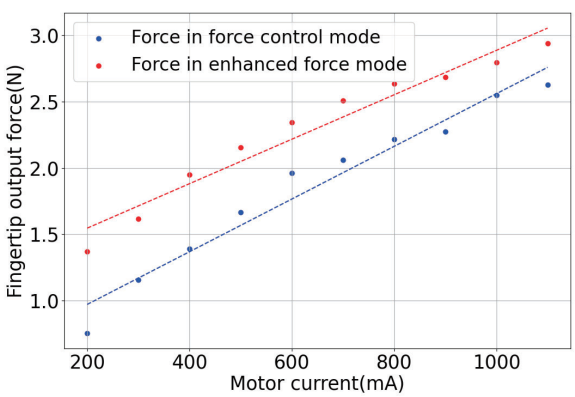

Fig. 7: Relationship between input current value and fingertip output force in enhanced force mode and force control mode.

(Fig. 4 (A)). Measurements were performed under three conditions: force applied solely at the first joint, solely at the second joint, and simultaneously at both joints (Fig. 6). The collected data were fitted with linear models, yielding correlation coefficients of 0.961, 0.919, and 0.967, respectively. These results demonstrate the feasibility of force control.

When force was applied solely at the first joint, the fingertip force was measured under both the force control mode and the enhanced force mode. The results are shown in Fig. 7. It can be observed that when both servo motors simultaneously actuate the syringe to drive a single joint, the available output force is significantly increased, demonstrating the effectiveness of the enhanced force mode in enhancing the actuation capability.

C. Demonstration of Object Grasping

To demonstrate HydroGrasp's adaptability in grasping diverse objects, we mounted the gripper on an XArm 6 robot and conducted a series of pick-and-place tasks. To evaluate its performance, a variety of objects were placed on a tabletop. The robot was programmed to approach, grasp, lift, and release each object. The selected items were chosen to assess the gripper's ability to handle objects with varying sizes, weights, shapes, and fragilities. The test set included deformable rubber toys, a rigid cup, and plastic fruit models. HydroGrasp successfully grasped all objects, demonstrating its versatility in handling a wide range of items. The resulting grasps are shown in Fig. 8.

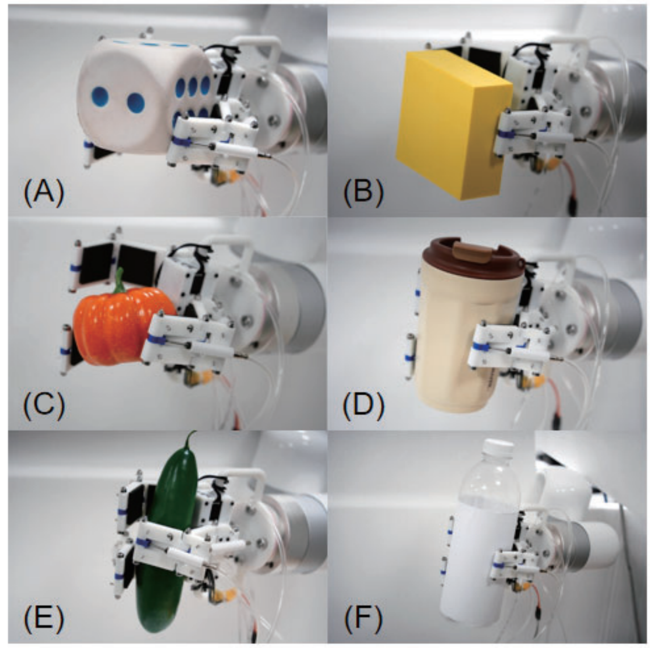

Fig. 8: Demonstration of manipulator's usability in grasping different daily life objects.

V. Discussions and Limitations

Overall, the experimental results demonstrate that HydroGrasp is a lightweight yet powerful robotic manipulator. Its hybrid actuation system effectively decouples actuator mass from the end-effector, enabling the transmission of large forces to the fingers from a remote location. However, the current design has several limitations that suggest directions for future improvement. First, a common limitation of simple syringe-based hydraulic systems is slow retraction speed, as fluid return often relies on passive forces from atmospheric pressure and may result in vacuum formation (cavitation) during high-speed movements. To accelerate retraction when needed, the design can optionally incorporate extension springs at each finger joint, which actively push the fluid back into the syringe. While the use of antagonistic springs for joint retraction is effective, it introduces a constant resistive load that the actuation system must overcome. This slightly reduces the net output force and increases energy consumption. Second, the system currently relies on commercially available pneumatic cylinders, as hydraulic cylinders of comparable size and weight are not readily accessible. Although robust, these components were not originally designed for hydraulic fluids.

Future work will focus on designing and fabricating custom, compact hydraulic cylinders with integrated fluid return channels or built-in springs, eliminating the need for external springs and improving overall efficiency and compactness.

In addition, we plan to integrate fingertip tactile sensors to enable closed-loop force control, thereby allowing the gripper to perform more delicate and precise manipulation tasks.

VI. Conclusions

In this paper, we presented HydroGrasp, a novel three-fingered robotic gripper that leverages a hybrid electrical-hydraulic actuation system to achieve a high force-to-weight ratio. By relocating the primary actuators to a remote base via hydraulic transmission, the system significantly reduces distal inertia while enabling high force output. HydroGrasp supports both position and force control across different dynamic ranges, enhancing its adaptability for a wide variety of manipulation tasks.

Experimental evaluations validated the design, demonstrating a peak fingertip force of 3.38 N under force control mode and a payload capacity of 0.675 kg under enhanced force mode. We further showcased the gripper's versatility through successful grasps of diverse objects. By constructing the system from accessible, low-cost components, HydroGrasp provides a powerful and replicable platform for the robotics community. Future work will focus on developing compact hydraulic cylinders and integrating tactile sensors.

Appendix

The forward kinematics of a single finger is represented by the homogeneous transformation matrix T. It describes the pose of the fingertip frame relative to the base frame. The matrix is derived from the Denavit–Hartenberg (D-H) parameters listed in Table I.

$$T = T_1^0 \cdot T_2^1 \cdot T_3^2 = \begin{bmatrix} D_{11} & D_{12} & -\sin\theta_1 & D_{14} \\ D_{21} & D_{22} & \cos\theta_1 & D_{24} \\ D_{31} & D_{32} & 0 & D_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

where the elements D_{ij} are defined as:

$$\begin{cases} D_{11} = \cos\theta_1 \cos(\theta_2+\theta_3) \\ D_{12} = -\cos\theta_1 \sin(\theta_2+\theta_3) \\ D_{14} = \cos\theta_1(a_1 + a_2\cos\theta_2 + a_3\cos(\theta_2+\theta_3)) - d_1\sin\theta_1 \\ D_{21} = \sin\theta_1 \cos(\theta_2+\theta_3) \\ D_{22} = -\sin\theta_1 \sin(\theta_2+\theta_3) \\ D_{24} = \sin\theta_1(a_1 + a_2\cos\theta_2 + a_3\cos(\theta_2+\theta_3)) + d_1\cos\theta_1 \\ D_{31} = \sin(\theta_2+\theta_3) \\ D_{32} = \cos(\theta_2+\theta_3) \\ D_{34} = a_2\sin\theta_2 + a_3\sin(\theta_2+\theta_3) + d_1 \end{cases}$$

References

[1] J. T. Belter and A. M. Dollar, "Performance characteristics of anthropomorphic prosthetic hands," in *2011 IEEE International Conference on Rehabilitation Robotics*, 2011, pp. 1–7.

[2] J. N. Ingram, K. P. Körding, I. S. Howard, and D. M. Wolpert, "The statistics of natural hand movements," *Experimental brain research*, vol. 188, no. 2, pp. 223–236, 2008.

[3] Shadow Robot Company. (2021) Shadow dexterous hand technical specification. [Online]. Available: https://www.shadowrobot.com/wp-content/uploads/2022/03/shadow_dexterous_hand_e_technical_specification.pdf

[4] R. Peng, Y. Wang, M. Lu, and P. Lu, "A dexterous and compliant aerial continuum manipulator for cluttered and constrained environments," *Nature Communications*, vol. 16, no. 1, p. 889, 2025.

[5] H. Yin, S. Huang, M. He, and J. Li, "An overall structure optimization for a light-weight robotic arm," in *2016 IEEE 11th conference on industrial electronics and applications (ICIEA)*. IEEE, 2016, pp. 1765–1770.

[6] C. Xiao and J. Wachs, "Complacent: A compliant whisker manipulator for object tactile exploration," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2023, pp. 10 184–10 190.

[7] L. Jiang, C. Fu, Y. Liang, Y. Jin, and H. Wang, "Critical review on the relationship between design variables and performance of dexterous hands: a quantitative analysis," *Frontiers in Neurorobotics*, vol. 18, p. 1513458, 2025.

[8] C. B. Pham, S. H. Yeo, and G. Yang, "Tension analysis of cable-driven parallel mechanisms," in *2005 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2005, pp. 257–262.

[9] X. Yin and A. Bowling, "Study of effect of cable yielding in cable-driven robotic system," in *IEEE International Conference on Robotics and Automation, 2004. Proceedings. ICRA'04. 2004*, vol. 4. IEEE, 2004, pp. 3234–3239.

[10] J. Mattila, J. Koivumäki, D. G. Caldwell, and C. Semini, "A survey on control of hydraulic robotic manipulators with projection to future trends," *iEeE/ASME Transactions on Mechatronics*, vol. 22, no. 2, pp. 669–680, 2017.

[11] J. Zhou, J. Huang, Q. Dou, P. Abeel, and Y. Liu, "A dexterous and compliant (dexco) hand based on soft hydraulic actuation for human inspired fine in-hand manipulation," *IEEE Transactions on Robotics*, 2024.

[12] J. Qi, X. Li, Z. Tao, H. Feng, and Y. Fu, "Design and control of a hydraulic driven robotic gripper," in *2021 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, 2021, pp. 398–404.

[13] T. Kang, H. Kaminaga, and Y. Nakamura, "A robot hand driven by hydraulic cluster actuators," in *2014 IEEE-RAS International Conference on Humanoid Robots*, 2014, pp. 39–44.

[14] Z. Neusser, M. Nečas, J. Pelikán, J. Karlíček, V. Pawlik, P. Beneš, M. Machálka, Z. Šika, and M. Valášek, "Mechatronic stiffness of cable-driven mechanisms: a study on production machine model," *The International Journal of Advanced Manufacturing Technology*, vol. 123, no. 1, pp. 431–446, 2022.

[15] H. Asada and K. Youcef-Toumi, *Direct-drive robots: theory and practice*. The MIT Press, 1987.

[16] H. Xiong and X. Diao, "A review of cable-driven rehabilitation devices," *Disability and Rehabilitation: Assistive Technology*, vol. 15, no. 8, pp. 885–897, 2020.

[17] K. Shaw, A. Agarwal, and D. Pathak, "Leap hand: Low-cost, efficient, and anthropomorphic hand for robot learning," *arXiv preprint arXiv:2309.06440*, 2023.

[18] C. Borst, M. Fischer, S. Haidacher, H. Liu, and G. Hirzinger, "Dlr hand ii: Experiments and experience with an anthropomorphic hand," in *2003 IEEE international conference on robotics and automation (Cat. No. 03CH37422)*, vol. 1. IEEE, 2003, pp. 702–707.

[19] P. Probst, "A review of the role of robotics in surgery: To davinci and beyond!" *Missouri medicine*, vol. 120, no. 5, p. 389, 2023.

[20] M. Miyasaka, M. Haghighipanah, Y. Li, J. Matheson, A. Lewis, and B. Hannaford, "Modeling cable-driven robot with hysteresis and cable–pulley network friction," *IEEE/ASME Transactions on Mechatronics*, vol. 25, no. 2, pp. 1095–1104, 2020.

[21] J. Davies, P. T. Phan, D. Huang, T. T. Hoang, H. Low, M. T. Thai, C. C. Nguyen, E. Nicotra, N. H. Lovell, and T. N. Do, "Hydraulically actuated soft tubular gripper," in *2022 International Conference on Robotics and Automation (ICRA)*. IEEE, 2022, pp. 6144–6150.

[22] G. Nelson, A. Saunders, and R. Playter, "The petman and atlas robots at boston dynamics," in *Humanoid robotics: A reference*. Springer, 2017, pp. 1–18.

[23] U. Kim, D. Jung, H. Jeong, J. Park, H.-M. Jung, J. Cheong, H. R. Choi, H. Do, and C. Park, "Integrated linkage-driven dexterous anthropomorphic robotic hand," *Nature communications*, vol. 12, no. 1, p. 7177, 2021.